

Practice question – Equivalence partitioning and Boundary Value Analysis

For examination system, the result of students are being generated according to the following table.

Marks Secured	Letter Grade	Grade Point	Description of Performance
$\geq 91\%$	S	10	Outstanding
$\geq 86\%$ and $\leq 90\%$	A+	9	Excellent
$\geq 75\%$ and $\leq 85\%$	A	8	Very Good
$\geq 66\%$ and $\leq 74\%$	B	7	Good
$\geq 55\%$ and $\leq 65\%$	C	6	Satisfactory
$\geq 50\%$ and $\leq 54\%$	D	5	Pass
$< 50\%$	F	2	Fail
	E	2	Exposure (Absent)

Based on the equivalence partitioning technique, **show** all valid and invalid partitions and **write** test cases using equivalence partitioning.

Sample Answer:

Valid and Invalid Partitions:

Partition	Type
$\{101, \dots, \infty\}$	Invalid
$\{91, \dots, 100\}$	Valid
$\{86, \dots, 90\}$	Valid
$\{75, \dots, 85\}$	Valid
$\{66, \dots, 74\}$	Valid
$\{55, \dots, 65\}$	Valid
$\{50, \dots, 54\}$	Valid
$\{0, \dots, 49\}$	Valid
$\{-1, \dots, -\infty\}$	Invalid

Sample Test cases using equivalence Partitions:

Partition	Test Data (Student Score)	Expected Output
$\{101, \dots, \infty\}$	101	Shows error
$\{91, \dots, 100\}$	91	Letter Grade: S Grade Point: 10 Description of Performance: Outstanding
$\{86, \dots, 90\}$	86	Letter Grade: A+ Grade Point: 9 Description of Performance: Excellent
$\{75, \dots, 85\}$	75	Letter Grade: A Grade Point: 8 Description of Performance: Very Good
$\{66, \dots, 74\}$	66	Letter Grade: B Grade Point: 7 Description of Performance: Good
$\{55, \dots, 65\}$	55	Letter Grade: C Grade Point: 6 Description of Performance: Satisfactory

$\{50, \dots, 54\}$	50	Letter Grade: D Grade Point: 5 Description of Performance: Pass
$\{0, \dots, 49\}$	0	Letter Grade: F Grade Point: 2 Description of Performance: Fail
$\{-1, \dots, -\infty\}$	-1	Shows error